



GE Medical Systems

Technical Publications

Direction 2241394

Revision 4

Signa[®] OpenSpeed Field Replaceable Units (FRU)

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GE Medical Systems

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**COVER
BOTTOM
CROP LINE**

**COVER
BOTTOM
CROP LINE**

**5/16" RADIUS
CORNER TRIM
TYPICAL**



**COVER
LEFT
CROP LINE**

**COVER
FOLD
LINE**

**(2) CENTER STAPLES
FOR BINDING**

**COVER
RIGHT
CROP LINE**



**5/16" RADIUS
CORNER TRIM
TYPICAL**

REVISION HISTORY

REV	DATE	REASON FOR CHANGE
0	OCT 20, '00	Initial Production Release
1	MAR 23, '01	Updated to match current product changes and omissions. Added sections for; 1) <i>Magnet Enclosure,</i> <i>Covers & Touch-up Paints,</i> Page 5. 2) <i>Main Disconnect Panel, Fuses,</i> Page 29.
2	NOV 22, '01	Updated to current product. Added Sections for; 1) <i>Workspace Scim And Keyboards</i> Page 15. 2) <i>Octane II Host Computer</i> Page 17.
3	DEC 23, '02	Updated to current product. Pay close attention to Sections; <i>Octane I & II Host Computers,</i> Pages 16 & 17.
4	MAR 18, '03	Updated to current product. 1) Updates to <i>Octane II</i> Illustration Page 17. 2) Added FRUs to <i>Power Cab.</i> , Pages 20& 21. 3) Updates to <i>Body Flex Coils</i> Page 27. 4) Added MCR FRUs to <i>Service Tools,</i> Page 25.

SEND COMMENTS ABOUT THIS MANUAL TO:

GE MEDICAL SYSTEMS
MANAGER, SERVICE ENGINEERING
W-808
3200 N. GRANDVIEW BLVD.
WAUKESHA, WISCONSIN 53188
USA

This manual belongs to:

INTRODUCTION TO FIELD REPLACEABLE UNIT (FRU)

This direction is a guide to the FRU's.

Parts designated as a FRU are critical to maintain the safety and/or functionality of the system. FRU's are available on a Priority 1 basis from Service Parts Operation.

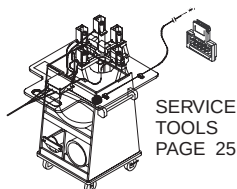
As a reference aid, the FRU's are listed under an upper-level part number where they are used. In most cases the upper-level part number is not a FRU. In some cases the parts listed under an upper-level assembly may have been nested in intermediate sub-assemblies which are not shown.

Where possible, FRU's which are part of Option Catalogs are shown under the area where they would be installed, rather than being listed in the "Coils & Options" section, Pages 26 & 27.

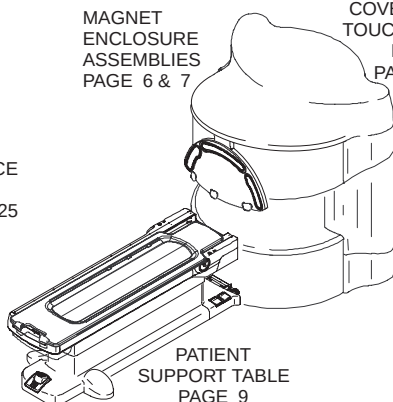
TITLE	PAGE
INTRODUCTION TO FIELD REPLACEABLE UNIT(FRU) . . .	1
MAGNET ENCLOSURE (COVERS & TOUCH-UP PAINTS) . .	5
MAGNET ENCLOSURE (ASSEMBLIES)	6 & 7
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SERVICE TOOLS	25
COILS & OPTIONS	26 & 27
MAIN DISCONNECT PANEL (FUSES)	29

MAGNET
ENCLOSURE
ASSEMBLIES
PAGE 6 & 7

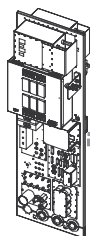
COVERS &
TOUCH-UP
PAINT
PAGE 5



SERVICE
TOOLS
PAGE 25



PATIENT
SUPPORT TABLE
PAGE 9



PENETRATION
PANEL
PAGE 11

MAGNET ROOM

OPERATOR ROOM

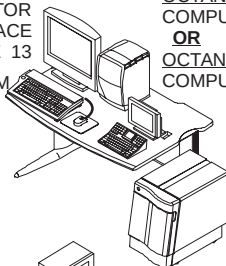
OPERATOR
WORKSPACE
PAGE 13

OCTANE HOST
COMPUTER PAGE 16

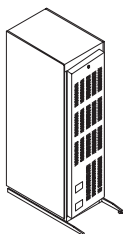
OR

OCTANE II HOST
COMPUTER PAGE 17

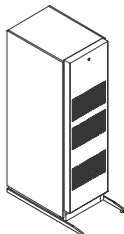
WORKSPACE SCIM
AND KEYBOARDS
PAGE 15



OPERATOR
WORKSPACE
CABINET
PAGE 19



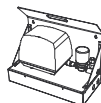
POWER
CABINET
PAGES
20 & 21



SYSTEM
CONTROL
CABINET
PAGE 23



MAIN
DISCONNECT
PANEL(FUSES)
PAGE 29



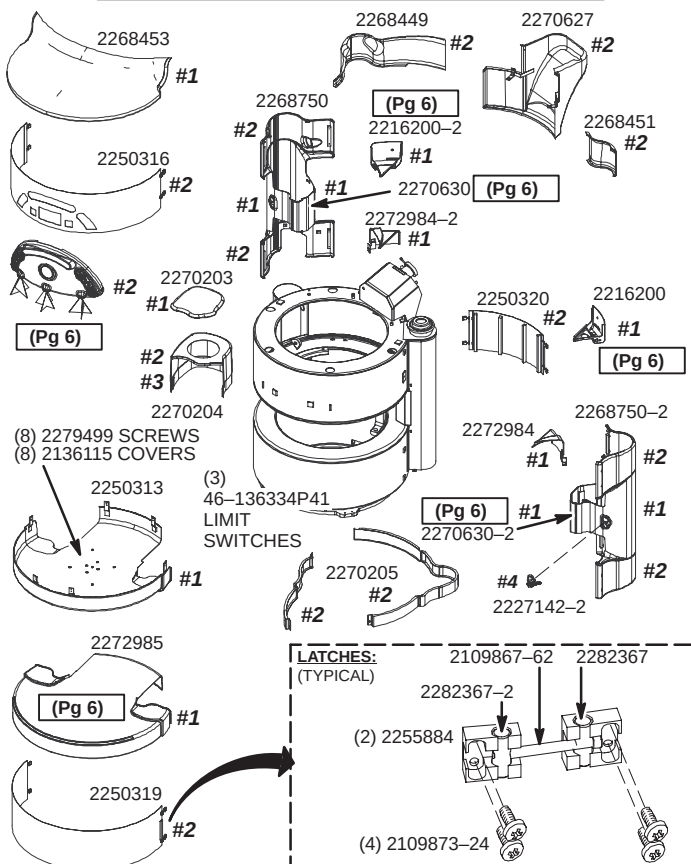
PATIENT
AIR SUPPLY
COMPRESSOR
PAGE 7

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This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

TOUCH-UP PAINTS:

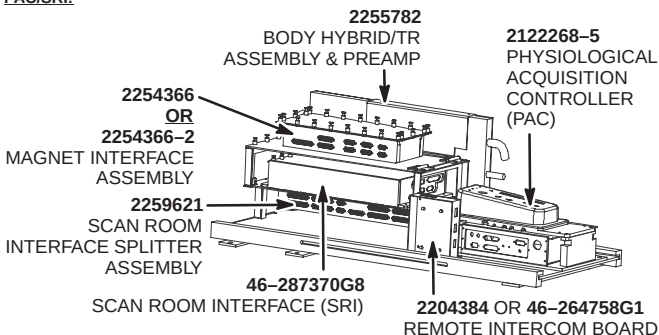
2183491-3	GEMS GRAY #1	0.6 OZ. Non-Aerosol
2183491	GEMS GRAY #2	0.6 OZ. Non-Aerosol
2183492	GEMS GRAY #2	4.5 OZ. Aerosol
2183491-2	GEMS GRAY #3	0.6 OZ. Non-Aerosol
2183492-2	GEMS GRAY #3	4.5 OZ. Aerosol
2180026	GEMS GRAY #4	0.6 OZ. Non-Aerosol



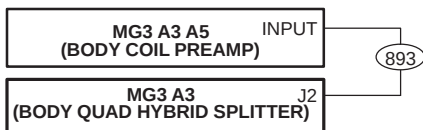
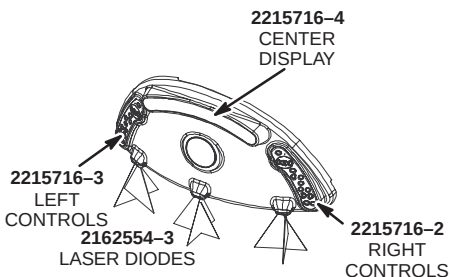
NOTE: COVER PART NUMBERS FOR REFERENCE AND ALL MAY NOT BE FRUS.

REVISION 4

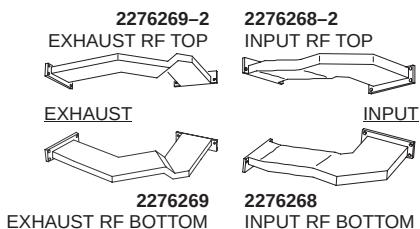
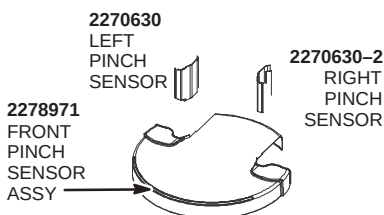
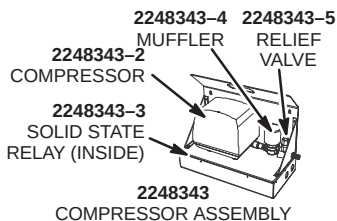
DIRECTION 2241394

PAC/SRI:CABLE:

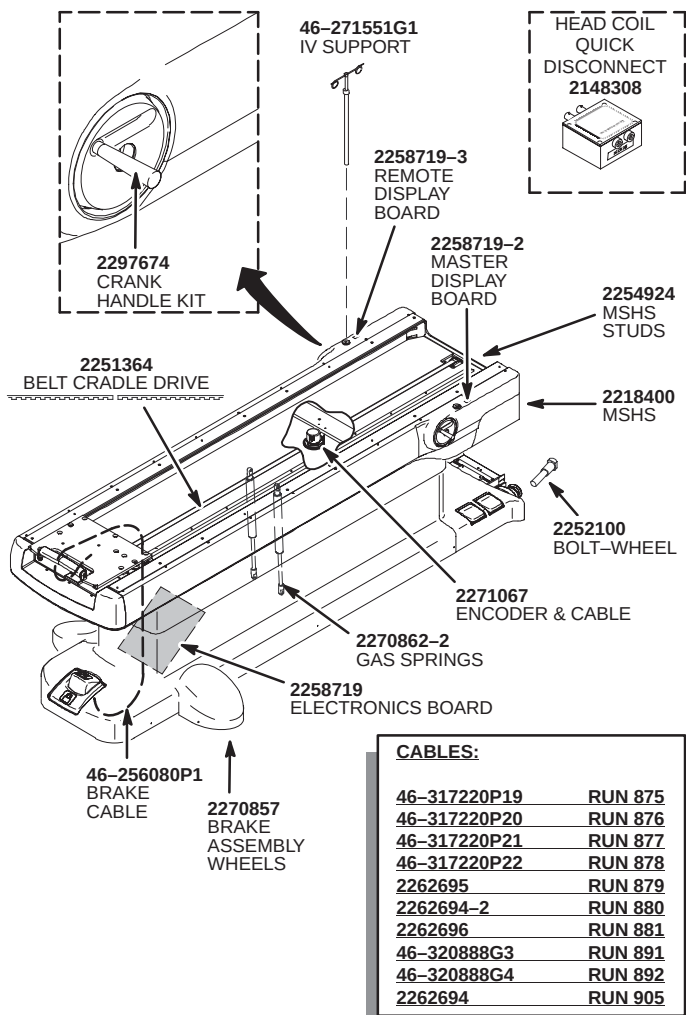
2364795 RUN 893 MG3 A3 J2 TO MG3 A3 A5 IN

DISPLAY/CONTROLS:LAMPS:

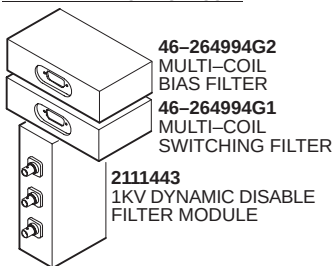
2257458
HALOGEN LAMP
12V20W
Q20MR16C/CG40DE

AIR DUCTS: VIEWED FROM FRONT OF MAGNET**PINCH SENSORS:****COMPRESSOR:**

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



[illegible]

FILTER PLATE FOR MULTICOIL:**PARTS INTERNAL TO THE 2222016
THERMAL CONTROL UNIT (TCU):**

46-302200P6 RATING PLATE
(NOT SHOWN)

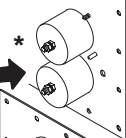
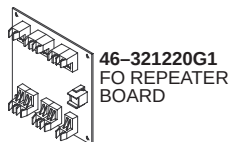
46-320232G1
GRADIENT
FILTER ASM

2132620
CE MARKING
LABEL CLASS A

2222016
NMR THERMAL
CONTROL UNIT

46-302200P6
LABEL
(FAR SIDE OF TCU)

46-271110P1
LABEL (GROUND)
PROTECTIVE EARTH



46-307169P1
1/4 NPT, PARKER
#207ACBH-4



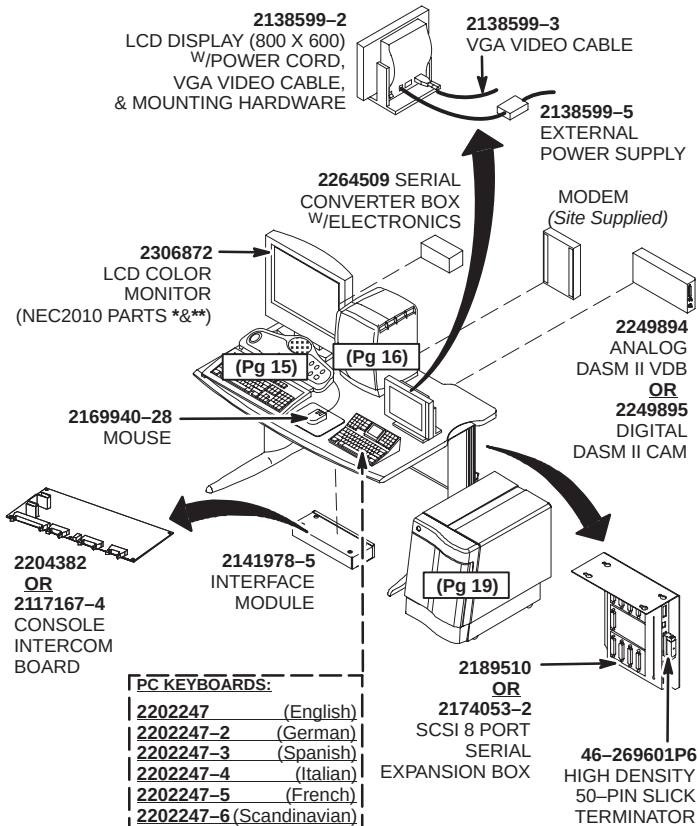
46-271110P1
LABEL (GROUND)
PROTECTIVE EARTH

NOTE:

* **46-282750P1**
250VAC/1000VDC EMI NOISE FILTER
(HARDWARE INCLUDED)

PENETRATION PANEL

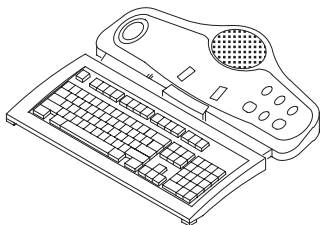
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



NOTE: The following parts are **Not Shown**.

2174053-3	SPARE SERIAL EXPANDER POWER SUPPLY
2156158-32	S/W COLLECTOR HFO II M4
2256092-3	SIGNA HORIZON LX IRIX 6.5 OS
2274462	8.3 M5 PATCH CD-ROM
2160200-5	CPD MODULE
*2213962-4	POWER SUPPLY FOR NEC2010 MONITOR
**2150533-2	CABLE, NEC2010 MONITOR (821/798) OW1A15/OW1A2A3

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

LATEST VERSIONS:**KEYBOARDS ONLY:**

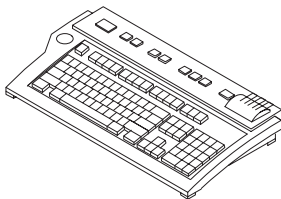
2275756 (English)
*2275756-2 (French)
*2275756-3 (German)
*2275756-4 (Scandinavian)
*2275756-5 (Italian)
*2275756-6 (Portuguese)
*2275756-7 (Spanish)

*** NOTE:**

These types of Keyboard/SCIM Assemblies were not available at the time this book was released, see below for earlier types.

NOTE: The following parts are Not Shown.

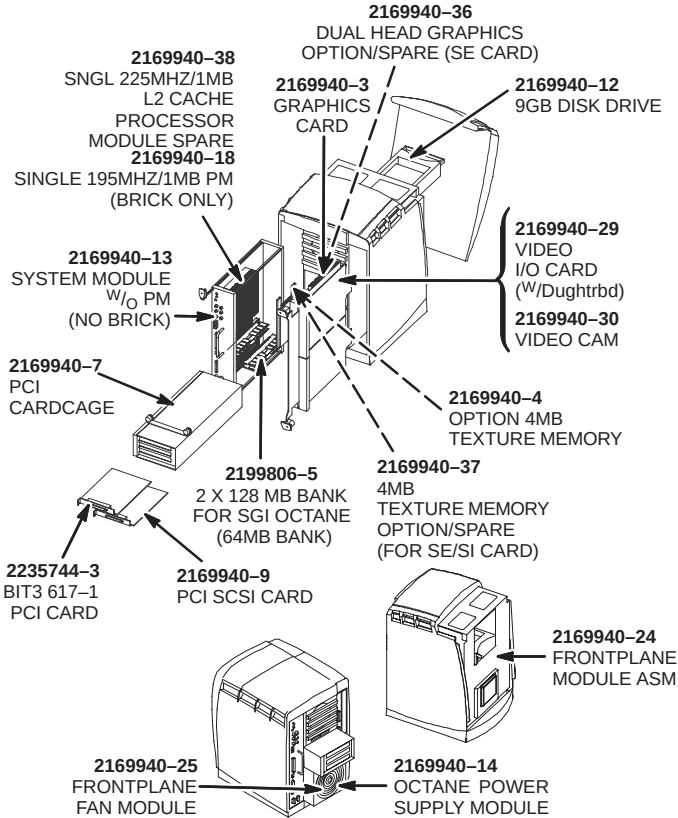
2262158-9	CABLE, 50 PIN SCIM KEYBOARD
2289697-2	SCIM ^W /ENGLISH OVERLAY
2289697-8	MR SCIM GLOBAL OVERLAY SET

EARLY VERSIONS:**KEYBOARDS ONLY:**

2114561-19 (English)
2114561-28 (French)
2114561-31 (German)
2114561-34 (Scandinavian)
2114561-43 (Italian)
2114561-44 (Portuguese)
2114561-45 (Spanish)

NOTE: The following part is Not Shown.

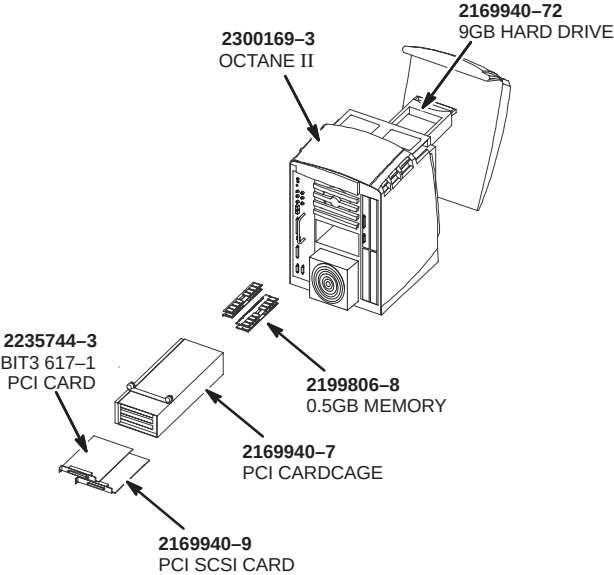
2114561-46	CABLE, LX SCSI LONG CABLE
------------	---------------------------



NOTE: The following parts are **Not Shown**.

2169940-78	18GB DISK DRIVE
2169940-32	SPARE LIGHTBAR ASSEMBLY
2169940-33	SGI MICROPHONE
2295019	CABLE, VGA (LCD MONITOR) to 13W3-DB15 RGB
2256485	CABLE, 3BNC-F to HD15 RGB (CRT MONITOR)
2169940-6	OPTION SINGLE 128MB MEMORY MODULE
2306872	COLOR LCD (HOST) MONITOR
2235744-4	IMAGE TRANSFER-HOST BOARD

-WARNING-
DO NOT INTERCHANGE FRU
PARTS BETWEEN OCTANE I
AND OCTANE II COMPUTERS.

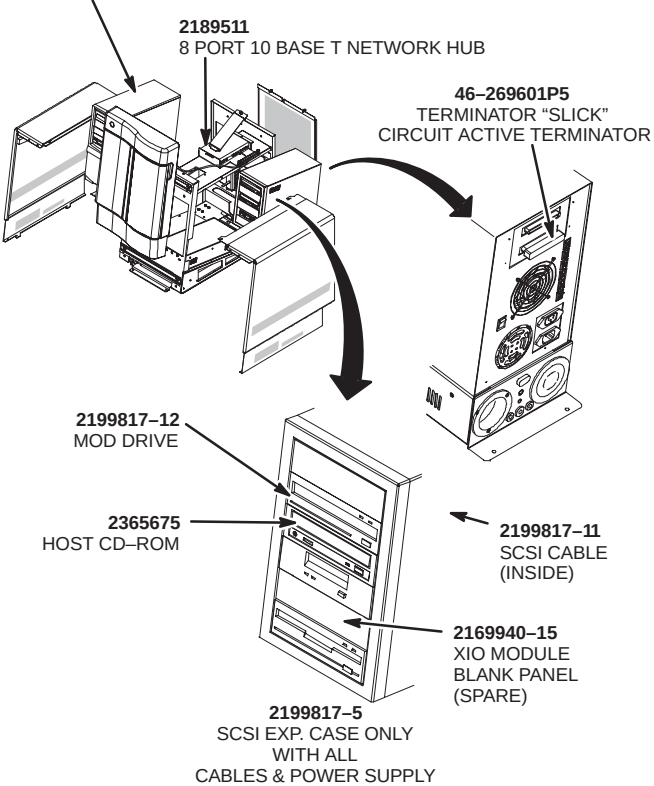


NOTE: The following parts are Not Shown.

2169940-78	18GB DISK DRIVE
2169940-32	SPARE LIGHTBAR ASSEMBLY
2169940-33	SGI MICROPHONE
2295019	CABLE, VGA (LCD MONITOR) to 13W3-DB15 RGB
2256485	CABLE, 3BNC-F to HD15 RGB (CRT MONITOR)
2169940-6	OPTION 128MB MEMORY MODULE
2306872	COLOR LCD (HOST) MONITOR
2235744-4	IMAGE TRANSFER-HOST BOARD

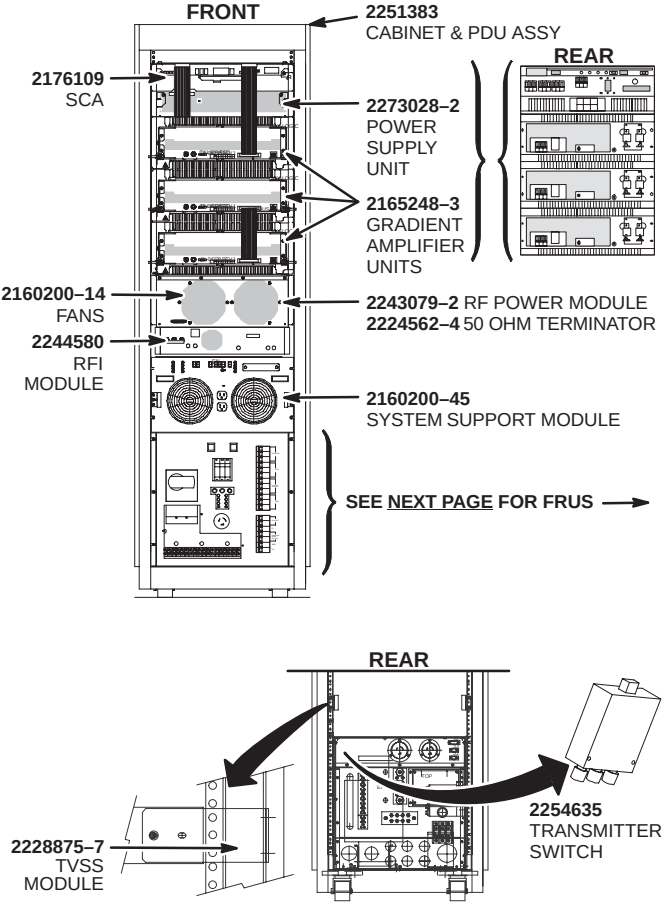
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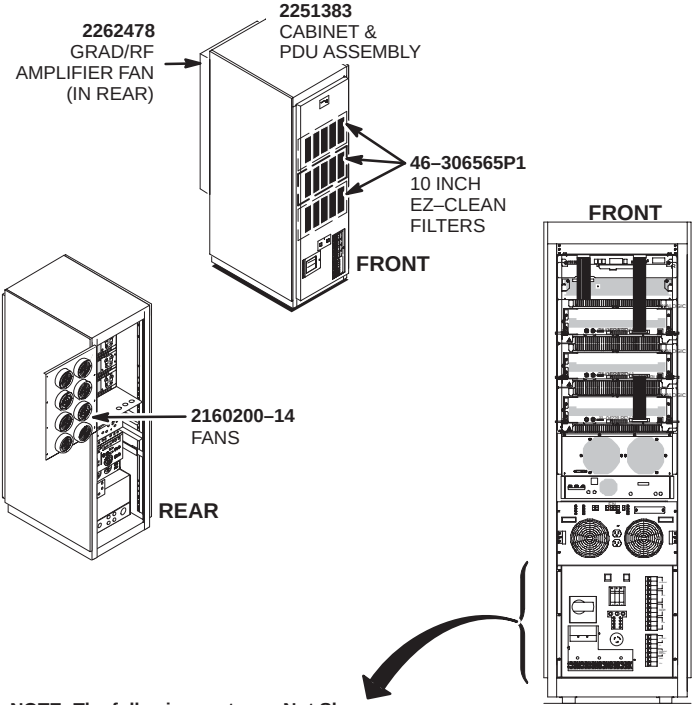
- 2284297 BOOT FLOPPY
- OR
- 2192746-3 PC WORKSTATION LX3 BASE SYSTEM WITH 2192746-4 S/W
- OR
- 2192746 PC WORKSTATION LX2 PC WORKSTATION WITH MOD



NOTE: The following parts are Not Shown.

2169940-21	XIO CONNECTOR PROTECTIVE CAP
2169940-32	LIGHTBAR MODULE ASM
2169940-33	SGI MICROPHONE
2199817-7	ALT. SCSI EXPANSION BOX HOST CD-ROM

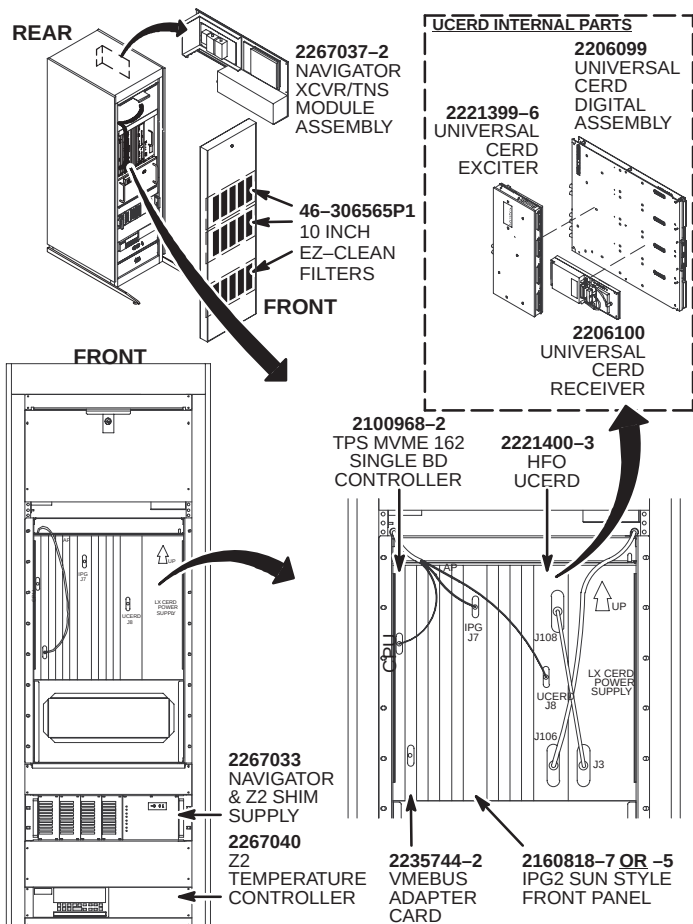




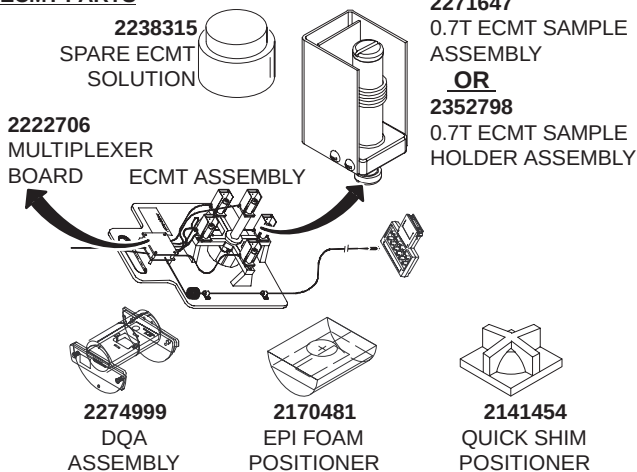
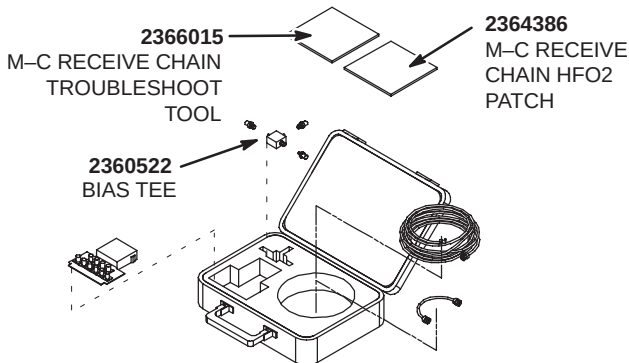
NOTE: The following parts are Not Shown.

2228875-2	CONTROL PC BOARD
2228875-3	EMERGENCY RESET/POWER SWITCH
2228875-6	INPUT CIRCUIT BREAKER MODULE
2228875-7	TVSS MODULE
2228875-8	GREEN NEON INDICATOR
2228875-9	CIRCUIT BREAKER 16A 1 POLE
2228875-10	CIRCUIT BREAKER 20A 1 POLE
2228875-11	CIRCUIT BREAKER 20A 2 POLE
2228875-12	CIRCUIT BREAKER 30A 3 POLE
2228875-13	CIRCUIT BREAKER 60A 3 POLE
2228875-14	CIRCUIT BREAKER 30A 1 POLE
2228875-15	SWITCH, SPDT, 20A

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[illegible]

ECMT PARTS**MCR; 2360581, 0.7T M-C RECEIVE TOOL KIT**

NOTE: Where possible, FRU's which are part of Option Catalogs are shown under the area where they would be installed, rather than being listed in this section.

HEAD COIL (LARGE BASEPLATE) M2087H

<u>2274240-7</u>	HEAD ARRAY COIL & BASEPLATE ASM ^{W/} INTRFC CABLE
<u>2274240-2</u>	HEAD ARRAY COIL SYSTEM INTERFACE CABLE
<u>2274240-5</u>	HEAD ARRAY PHANTOM
<u>2274240-8</u>	HEAD ARRAY PHANTOM POSITIONER

HEAD COIL (SMALL BASEPLATE) M2087HA

<u>2299522-2</u>	HEAD ARRAY COIL & BASEPLATE ASM ^{W/} INTRFC CABLE
<u>2274240-2</u>	HEAD ARRAY COIL SYSTEM INTERFACE CABLE
<u>2274240-5</u>	HEAD ARRAY PHANTOM
<u>2274240-8</u>	HEAD ARRAY PHANTOM POSITIONER

BREAST COIL M2087BR

<u>2264745-7</u>	BREAST COIL ^{W/} CABLE
<u>2264745-2</u>	BREAST COIL CABLE ONLY
<u>2246360-9</u>	PHANTOM POSITIONER FOR BREAST COIL
<u>46-317586G1</u>	SPHERE PHANTOM

CTL COIL (SPINE) M2087HS

<u>2264744-12</u>	CTL ARRAY ^{W/} CABLE
<u>2264744-2</u>	CTL CABLE ASSEMBLY ^{W/} QD BOX
<u>2264744-11</u>	CTL ARRAY LATCH MECHANICS KIT
<u>2264744-7</u>	CTL ARRAY CT PHANTOM
<u>2264744-8</u>	CTL ARRAY TL PHANTOM

SHOULDER COIL M2087SH

<u>2264743-9</u>	SHOULDER ARRAY COIL ^{W/} CABLE
<u>2264743-2</u>	SHOULDER ARRAY COIL CABLE ASSEMBLY
<u>2264743-5</u>	SHOULDER ARRAY PHANTOM POSITIONER
<u>46-265826G6</u>	HEAD TLT PHANTOM

EXTREMITY / FOOT SET M2087E

<u>2264742-2</u>	LARGE KNEE COIL ^{W/} CABLE ASSEMBLY
<u>2264742-14</u>	LARGE KNEE COIL CABLE ASSEMBLY
<u>2264742-8</u>	KNEE COIL PHANTOM (LARGE OR SMALL)
<u>2264742-10</u>	PHANTOM POSITIONER (LARGE & SMALL)
<u>2264742-3</u>	SMALL KNEE AND FOOT COIL ^{W/} CABLE
<u>2264742-5</u>	SMALL KNEE / FOOT COIL CABLE ASSEMBLY
<u>2264742-13</u>	LATCH PARTS (SET OF 2)
<u>2264742-4</u>	FOOT PHANTOM

WRIST COIL M2087WR

<u>2279915-2</u>	PHASED ARRAY WRIST COIL & BASEPLATE ASSEMBLY
<u>2279915-11</u>	PHASED ARRAY WRIST COIL BASEPLATE CABLE
<u>2279915-6</u>	PHASED ARRAY WRIST COIL PHANTOM
<u>2279915-10</u>	PHASED ARRAY WRIST COIL PHANTOM POSITIONER

COILS CONTINUED ON NEXT PAGE.

*COILS CONTINUED FROM PREVIOUS PAGE.***LARGE BODY FLEX COIL** M20032GC2273180-2 BODY FLEX L COIL ^{W/} CABLE2148175 BODY FLEX LOCK M12148176 BODY FLEX LOCK M22169786-2 BODY FLEX LOCK F12169787-2 BODY FLEX LOCK F2**MEDIUM BODY FLEX COIL** M20022GC2273181-2 BODY FLEX M COIL ^{W/} CABLE2148175 BODY FLEX LOCK M12148176 BODY FLEX LOCK M22169786-2 BODY FLEX LOCK F12169787-2 BODY FLEX LOCK F2**INTEGRATED BODY COIL** M2085BG/M2085BH2269627-2* OR 2269627-4* NMR RF BODYCOIL ASSEMBLY, TOP2269627* OR 2269627-3* NMR RF BODYCOIL ASSEMBLY, BOTTOM

* NOTE: You cannot mix 2269627-2 (Top) and 2269627 (Bottom) with 2269627-4 (Top) and 2269627-3 (Bottom).

2267178** RIGHT DYNAMIC DISABLE BOARD2267180** LEFT DYNAMIC DISABLE BOARD

** NOTE: 2267178 and 2267180 may be used on 2269627-2 (Top), 2269627 (Bottom), 2269627-4 (Top), and 2269627-3 (Bottom).

OPTIONS:**AP POWER PC W/128MB – DUAL**2222532 PPC COMPATIBLE MEDCAM BASEBOARD2222532-3 DUAL PPC MODULE W/64MB (32MB/CE NODE)2101767-4 MEDCAM VMX 128 MB MEMORY BOARD2101767-5 MEDCAM VMX FOREPLANE 1-SLOT COVER

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

NOTE: Refer to the Directions on CD-2250758-1 (Signa OpenSpeed Service Methods) for additional product information.

TABLE 1
50 HZ PARTS

PART NUMBER	LOCATION	MANUFACTURER	
		BUSS	G-S
2276547-2	FUSE F1	FNQ-R-1-1/4	ATQR1-1/4
2276547-3	FUSE F2	FNQ-R-3	ATQR3
2276547-5	FUSE F3	FNQ-R-30	ATDQR30
2276547-6	FUSE F4	FNQ-R-8	ATDR8
2276547-8	FUSE F5	FNQ-R-1	ATDR1
2276547-9	FUSE F6	-----	ATQR1/8
2276547-4	FUSE F7	LPJ-40SP	AJT40
2276547-10	FUSE F8	LPJ-70SP	AJT70
2276547-12	FUSE F9	LPJ-15SP	AJT15
2276547-11	FUSE F10	FNQ-R-10	ATDR10
2276547-11	FUSE F11	FNQ-R-10	ATDR10
2276547-13	FUSE F12	FNQ-R-15	ATDR15

TABLE 2
60 HZ PARTS

PART NUMBER	LOCATION	MANUFACTURER	
		BUSS	G-S
2276547-2	FUSE F1	FNQ-R-1-1/4	ATDR1-1/4
2276547-3	FUSE F2	FNQ-R-3	ATDR3
2276547-5	FUSE F3	FNQ-R-30	ATDR30
2276547-6	FUSE F4	FRS-R-8	ATDR8
2276547-8	FUSE F5	LP-CC-1	ATDR1
2276547-9	FUSE F6	SNQ-R-1/8	ATQR1/8
2276547-4	FUSE F7	LPJ-40SP	AJT40
2276547-15	FUSE F8	LPJ-90SP	AJT90
2276547-12	FUSE F9	LPJ-15SP	AJT15
2276547-11	FUSE F10**	FNQ-R-10	ATDR10
2276547-11	FUSE F11**	FNQ-R-10	ATDR10
2276547-13	FUSE F12**	FNQ-R-15	ATDR15

****NOTE:** Early design of this panel may not have had these fuses.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

[illegible]

[illegible]

NOTES